AP-IA1 COUNTER — THE SETTER'S OWN BANK, SOLVED

Exam: Applied Physics 1 IA-1, Tue 27 Oct 2026, 9:40 AM. Format (S-VYASA template, same as maths today): Section A 5×3M all, Section B 3×5M choose (A)-or-(B), 30M, 2h, show reasoning, no calculator, g=9.8 m/s² unless told (ONE question used g=10 — check the paper!).

Paper-setter's DNA (from his contests): MCQ = precise definition; 2M = single-step calc; 5M = (a)(b)(c) clean multi-part; signature = "explain without calculating" + story context (airplane, ball-kick, ice-tray). Questions are RE-PICKED FROM HIS WORKSHEETS WITH NEW NUMBERS. Own every method below.

BLOCK 1 — MEASUREMENT (L1) — his worksheet Q1–Q11

Q1 (defn, 3M script). Importance of measurement in engineering.

REASON: Measurement = numerical value + unit. Engineering needs it for (i) accuracy/quality control (ii) comparison across teams (iii) manufacturing consistency/tolerances (iv) scientific communication (v) safety limits. Without standardized units → mismatches (NASA Mars Orbiter).

Q2. Fundamental vs derived + SI units of any 5 fundamental.

REASON: Fundamental = independent, cannot be defined via others (mass M, length L, time T, temperature Θ, electric current I, amount of substance N, luminous intensity J). Derived = defined from fundamentals via laws (velocity LT^{-1}, force MLT^{-2}). Examples: length → meter; derived: speed = distance/time.

His 5: pick M, L, T, Θ, I (or N, J).

Q3. Dimensional formulas — memorize cold:

(a) Velocity = LT^{-1} (b) Acceleration = LT^{-2} (c) Force = MLT^{-2} (d) Work = ML^2T^{-2} (e) Pressure = $ML^{-1}T^{-2}$

Bonus he wrote on student paper: Planck constant h = ML^2T^{-1}; gravitational constant G = $M^{-1}L^3T^{-2}$. Also know: momentum MLT^{-1}, energy ML^2T^{-2}, power ML^2T^{-3}, angular momentum ML^2T^{-1}, surface tension MT^{-2}, viscosity $ML^{-1}T^{-1}$, frequency T^{-1}.

Q4. Scientific notation + order of magnitude:

(a) 0.000456 = 4.56×10^{-4}, order 10^{-4} (b) 7,850,000 = 7.85×10^{6}, order 10^{7} ($7.85 \geq \sqrt{10} \approx 3.16$ rounds up)

(c) 299,792,458 = 2.99792458×10^{8}, order 10^{8} (d) 0.0000000082 = 8.2×10^{-9}, order 10^{-8}

RULE: order of magnitude = round mantissa to 1,10-power: if mantissa $\geq\sqrt{10}\approx3.16$ → up, else down.

Q5. Rod measured 25.60 cm, true 25.30 cm. (a) Absolute error (b) Relative (c) Percentage.

(a) $\Delta L = |25.60-25.30| = 0.30$ cm (b) relative = 0.30/25.30 = 0.0119 (c) %= 1.19%

(His student wrote true=25.50 by misreading → 0.10, 0.00392, 3.92% — DON'T copy student's numbers, follow the printed text.)

Q6. Disc r=(10.0±0.2) cm. % error in area $A=\pi r^2$.

$\Delta A/A = 2(\Delta r/r) = 2\times0.2/10 = 0.04$ → **4%**. (Power rule: exponent multiplies relative error.)

Q7. A=(50.0±0.5), B=(20.0±0.2), Z=A/B. Max absolute + relative error.

$\Delta Z/Z = \Delta A/A + \Delta B/B = 0.01+0.01 = 0.02$ → relative 2%. Z=2.5 → $\Delta Z = 0.05$ → **Z = 2.5 ± 0.05**.

Q8. $s = ut + \frac{1}{2}at^2$ dimensional check.

[s]=L; [ut]=$LT^{-1}\cdot T=L$; [at^2]=$LT^{-2}\cdot T^2=L$. L = L+L ✓. (Write "both terms have dimension L, equation dimensionally correct".)

Q9 (story, his signature!). SI vs Imperial teams → mission failure.

REASON: 1999 Mars Climate Orbiter — Lockheed Martin used pound-force-seconds (Imperial thrust data), NASA navigation used newton-seconds (SI). Mixed units in same equation → navigation model off by factor 4.45 → orbiter entered Mars atmosphere too low and disintegrated. Lessons: enforce SI-only pipelines; unit-tag every variable; automated unit-checking software; dimensional analysis as sanity check; standardized interfaces between teams.

Q10. Cube side (5.00±0.02) cm. (a) % error in volume (b) max absolute error.

$V=s^3$. (a) $\Delta V/V = 3\Delta s/s = 3\times0.02/5 = 0.012$ → **1.2%**. (b) $\Delta V = 3s^2\Delta s = 3\times25\times0.02$ = **1.5 cm^3** (or 1.2%×125=1.5).

Q11 (story!). Mars Climate Orbiter: what happened, which units, lessons — same as Q9 answer. (1999, lost ~$125M; thrust data in lbf·s vs required N·s.)

EXAM PREDICTION for Block 1: error-propagation 5M (Q6/Q7/Q10 type) + dimensional-check 2M + Mars story 3M "explain". Q3-style MCQ on random vs systematic (from contest Q3: unpredictable variations = RANDOM error).

BLOCK 4/6 — FRICTION + NLM (L5–L6) — his worksheet Q1–Q11, g=9.8

Q1. 5 kg, μs=0.40, max force without moving. $f_{s,max} = \mu s\cdot m\cdot g = 0.40\times5\times9.8$ = **19.6 N**.

Q2. 10 kg pulled 35 N, μk=0.20. (a) N=mg=98 N (b) f_k=19.6 N (c) F_{net}=35−19.6=15.4 N (d) a=15.4/10=**1.54 m/s^2**.

Q3. 6 kg, μs=0.50, μk=0.30, F=20 N. $f_{s,max}=0.5\times6\times9.8=29.4$ N > 20 → DOES NOT MOVE. Actual static friction = **20 N** (self-adjusting, equals applied).

Q4. 8 kg, μs=0.40, μk=0.25, force ramped from 0. (a) moves at 0.40×8×9.8=**31.36 N** (b) just after: f_k=0.25×8×9.8=**19.6 N** (c) a=(31.36−19.6)/8=**1.47 m/s^2**.

Q5. 5 kg, F=40 N at 30° ABOVE horizontal, μk=0.20. (a) Fx=40cos30=**34.64 N** (b) Fy=40sin30=**20 N** (c) N=mg−Fy=49−20=**29 N** (pull reduces normal!) (d) f_k=0.2×29=**5.8 N** (e) a=(34.64−5.8)/5=**5.77 m/s^2**.

Q6. 10 kg, F=50 N at 30° BELOW horizontal (push!), μk=0.25. (a) N=mg+Fsin30=98+25=**123 N** (push increases normal!) (b) f_k=0.25×123=**30.75 N** (c) F_{net}=50cos30−30.75=43.30−30.75=**12.55 N** (d) a=12.55/10=**1.26 m/s^2**.

Q7. 10 kg on 30° incline, μk=0.20, slides down. (a) $W_{\parallel}$=mg sin30=**49 N** (b) $W_{\perp}$=mg cos30=**84.87 N** (c) N=84.87 N (d) f_k=0.2×84.87=**16.97 N** (opposes slide, up the plane) (e) a=(49−16.97)/10=**3.20 m/s^2** down the plane.

Q8. 5 kg on 37° incline, μs=0.50, rest or slide? g=10, sin37=0.6, cos37=0.8. $W_{\parallel}$=5×10×0.6=30 N; $f_{s,max}$=0.5×5×10×0.8=20 N. 30>20 → **SLIDES**. (Test: tan37=0.75 > μs=0.50 → slides. General rule: slides iff tanθ > μs.)

Q9. 4 kg on 30° incline, F=30 N up-parallel, μk=0.20. N=4×9.8×cos30=33.95; f_k=6.79; $W_{\parallel}$=19.6. Net up = 30−19.6−6.79 = +3.61 → a = 3.61/4 = **0.90 m/s^2 UP the plane**.

Q10. 6 kg pulled up 30° incline by T=50 N, μk=0.15. Draw FBD (weight components, N, T up-parallel, f_k down-parallel). N=6×9.8×cos30=50.92; f_k=7.64; $W_{\parallel}$=29.4; a=(50−29.4−7.64)/6=**2.16 m/s^2** up.

Q11 (pulley!). 5 kg on rough table (μk=0.20) + 3 kg hanging, frictionless pulley. Whole-system method: $a = (m_2g - \mu k\, m_1\, g)/(m_1+m_2)$ = (29.4 − 9.8)/8 = **2.45 m/s^2**. $T = m_2(g-a)$ = 3×7.35 = **22.05 N** (check: T − f_k = m_1a → 22.05−9.8=12.25=5×2.45 ✓).

FBD SCRIPT (he draws these!): box → N up, mg down, T right, f_k left. On incline: mg sinθ down-parallel, mg cosθ into plane.

BLOCK 5/7 — CIRCULAR MOTION (L7) — his worksheet Q1–Q30

Q1. Car, r=50 m, v=10 m/s. ω = v/r = **0.2 rad/s**.

Q6. Wheel r=0.5 m at 120 rpm. (a) ω=120×2π/60=**4π ≈ 12.57 rad/s** (b) v=rω=0.5×4π=**2π ≈ 6.28 m/s**.

Q11. Torque, F=100 N at r=0.5 m, angle with radius 90°/60°/0°. τ=rF sinθ: (a) **50 N·m** (b) 100×0.5×sin60=**43.3 N·m** (c) sin0=0 → **0** (force along radius does no torque — same reason centripetal does no work).

Formulas cold: ω=v/r=2πf=2π/T; $a_c=v^2/r=r\omega^2$; v=rω; τ=rF sinθ; α=dω/dt; rotational kinematics mirror linear (θ↔x, ω↔v, α↔a); point on axis r=0 → v=0, a_c=0. rpm→rad/s: ×2π/60. Q27-type: "car moves, speed doubles → a_c quadruples (v^2)". Q19: linear velocity of object at radius r on rotating wheel = rω. Q20: difference between linear and angular velocity — dimension + frame (tangent vs rotation). Q28-30: (a) angular acceleration (b) linear velocity (c) radial/centripetal (d) tangential acceleration $a_t=r\alpha$ (e) total $a=\sqrt{(a_t^2+a_c^2)}$.

BLOCK 6 — WORK & ENERGY (L8) — his worksheet Q1–Q15 (from photo inventory)

Core pattern of his Qs: W=Fs cosθ; $KE=\frac{1}{2}mv^2$; $W_{net}=\Delta KE$ (work-energy theorem); $PE_g=mgh$; spring $PE=\frac{1}{2}kx^2$; power=W/t=Fv; conservative force (gravity, spring) → path-independent, closed loop W=0; friction → non-conservative, $W=-f_k d$, converts to heat.

Q13 pattern: 4 kg block pushed — (a) work done by applied force (b) work done by friction (c) net work (d) final KE (e) final speed [$W_{net}=\Delta KE=\frac{1}{2}mv^2$]. Q14: block mass m, energy remains unchanged → only conservative forces act / KE+PE constant. Q15: particle falls under gravity with friction → total energy NOT conserved, lost to heat = $f_k \cdot d$.

ZERO-WORK SCRIPT: force perpendicular to motion → W=0 (centripetal on circular path; normal force on flat road; gravity at constant height).

BLOCK 8 — COLLISIONS + MOMENTUM (L10) — his worksheet (Q1–15 across photos)

Two-equation machine (write every time):

(1) momentum: $m_1u_1+m_2u_2 = m_1v_1+m_2v_2$

(2) restitution: $v_2-v_1 = e(u_1-u_2)$ [separation speed = e × approach speed]

Q13. Two identical balls elastic, first 8 m/s, second at rest. e=1 identical masses → EXCHANGE velocities: v_1=0, v_2=8 m/s. (One-line theorem: elastic + equal mass + target at rest → swap.)

Q9. 3 kg at 6 m/s hits 2 kg at rest, e=0.4. (a) approach=6; separation=e×6=2.4. Eqs: $3v_1+2v_2=18$; $v_2-v_1=2.4$ → v_2=5.04, v_1=2.64 m/s. (b) separation speed=2.4 m/s (c) approach speed=6 m/s.

Stick-together type (m_074: "objects move together", e=0.5/0.6/0.4): perfectly inelastic → common velocity $v=(m_1u_1+m_2u_2)/(m_1+m_2)$; KE lost = $\frac{1}{2}\mu_{rel}\, u^2$ (reduced mass form) — show that KE is max lost when e=0.

Impulse: $J=F\Delta t=\Delta p=m(v-u)$. Q8 of momentum worksheet: (a) initial momentum=mu (b) final common momentum (c) change in momentum (d) impulse — all equal to J.

Q17 conceptual: net external force zero → (a) total momentum CONSTANT (conserved) (b) velocity of centre of mass constant/zero-change (c) acceleration of CM = 0. REASON script: $F_{ext}=dp/dt$; isolated system → p const.

Two-stage (notes m_106): explosion → internal forces, momentum conserved each stage: (m_A+m_B)v_before = m_A v_A + m_B v_B.

2D momentum (notes m_107): resolve into x/y; conserve separately: $m_1u_1x+m_2u_2x = m_1v_1x+m_2v_2x$, same for y; then combine $\sqrt{(vx^2+vy^2)}$.

BLOCK 7 — CENTRE OF MASS (L9) + HIS CONTEST-2 SUBJECTIVE

His Worksheet Q1–4 (m_104): two particles, three particles CoM.

$x_{CM}=(m_1x_1+m_2x_2+\ldots)/(m_1+m_2+\ldots)$; 2D → both x_CM, y_CM; $R_{CM}=\Sigma m_i r_i/M$. Symmetry shortcut: uniform rod → midpoint; uniform disc/cube/sphere → geometric centre; L-shaped → split into rectangles.

His Contest-2 question (0/40 XP, still open — STUDY THE ANSWER PATTERN): ice cube (m, L) melts on tray (M), water spreads thin. Does CoM change?

REASONING SCRIPT (his marking scheme wants this): CoM height depends on vertical distribution of mass. Before melt: ice CoM at L/2 above tray base, so combined CoM sits slightly above base ~ mL/2/(M+m). After melt: all mass lies in negligibly thin layer at tray base → water CoM at base level. Combined CoM drops toward the base. Magnitude of drop = [mL/2]/(M+m) — larger when m≫M, smaller when M≫m. Direction: always downward (mass redistributed toward base). No numbers, pure physical reasoning — EXACTLY his style: "without performing numerical calculations, explain... discuss how redistribution affects... relative masses influence magnitude."

Ice-tray TEMPLATE for exam variants: (i) does CM move? yes/no (ii) which direction (iii) what sets magnitude (mass ratio + height change).

BLOCK 2/3 — VECTORS, CALCULUS, KINEMATIC GRAPHS, PROJECTILE (L2–L4)

His contest Q13 (MCQ): vector × negative scalar → MAGNITUDE scales, DIRECTION REVERSED (option: direction is reversed).

His contest Q18: (1) dy/dx of $y=4x^2-3x+7 = 8x-3$ (2) $\int(3x^2+2x)dx = x^3+x^2+C$.

His contest Q19: position-time graphs: slope = velocity; straight line → constant velocity (avg = instantaneous, both constant); curved (parabola) → velocity changing (avg over interval ≠ instantaneous at point; instantaneous = slope of tangent).

His contest Q24: ball kicked 25 m/s at 37°: (1) ux=25cos37=20 m/s, uy=25sin37=15 m/s (2) $T=2uy/g=30/9.8\approx3.06$ s (or g=10 → 3.0 s) (3) $H=uy^2/(2g)=225/19.6\approx11.5$ m (g=10 → 11.25 m).

His contest Q28b: airplane 100 km/h pointing north, wind 60 km/h east. v_ground = vector sum: magnitude $\sqrt{(100^2+60^2)}=\sqrt{13600}\approx116.6$ km/h; direction $\theta=\tan^{-1}(60/100)\approx31°$ east of north. REASON: ground velocity = airspeed (through air) + wind velocity (air over ground).

Projectile master formulas: ux=u cosθ, uy=u sinθ; T=2uy/g; $H=uy^2/2g$; $R=u^2\sin2\theta/g$; at any time: vx=ux (constant), vy=uy−gt; angle of velocity: tanϕ=vy/vx. RANGE MAX at θ=45°; complementary angles (θ, 90−θ) → same range.

Relative motion (worksheet Q types m_050): v_AB = v_A − v_B; boat/river: time to cross = d/v_boat (perpendicular), drift = v_stream × t; rain-man umbrella angle: tanθ = v_rain/v_man.

Vector arithmetic: A·B=AB cosθ (scalar), A×B=AB sinθ (vector, right-hand); unit vector $\hat{A}=A/|A|$; resolution along x/y; triangle law + parallelogram law: $|A+B|=\sqrt{(A^2+B^2+2AB\cos\theta)}$.

DEFINITION FLASH — his 1M MCQs test THESE exact words

- Error from unpredictable/random variations = **Random error** (vs systematic = consistent bias, gross = human blunder)
- Vector × (–k): magnitude ×k, direction reversed
- Work by centripetal force on circular motion = **zero** (force ⊥ displacement)
- Static friction = self-adjusting, ≤ μsN; kinetic = constant μkN (usually μk < μs)
- Coefficient of restitution e = separation speed / approach speed; e=1 elastic, e=0 perfectly inelastic
- Momentum conserved when **net external force = 0** (internal forces cancel in pairs — Newton's 3rd)
- CoM of uniform symmetric body = geometric centre
- Order of magnitude: mantissa ≥√10 → round up the power
- Dimensions: M mass, L length, T time, Θ temp, I current, N amount, J luminous
- SI base units: kg, m, s, K, A, mol, cd
- Torque τ = rF sinθ; zero when force ∥ r
- Angular: ω=v/r, $a_c=v^2/r=r\omega^2$, $a_t=r\alpha$
- Normal force on horizontal surface = mg (no other vertical force); with F at angle above: mg−F sinθ; push below: mg+F sinθ

EXAM-DAY PROTOCOL (calibrated to today's maths pain)

1. FIRST pass (35 min): Section A all five — definitions flash + one-step calcs from this bank. Never blank: write the formula + definition sentence (partial marks exist; today's paper awarded (1)(1)(1) splits).

2. SECOND pass (60 min): Section B — pick the (A)/(B) pair matching THIS bank hardest (friction-incline pulley / collisions e / error propagation). Each 5M = (a)(b)(c) — label parts, show numbers+units.

3. LAST (15 min): the "explain" questions — use the REASONING SCRIPTS above verbatim-structure: claim → physical law → why → magnitude-influence. His CM ice-tray answer = the template.

4. If a question looks novel: it's a worksheet question with changed numbers. Identify block → pull THIS method → re-run numbers. The gaps that hurt maths (novelty panic) die here because you own his bank.

5. g: default 9.8; if the paper says g=10 (like friction Q8 did), switch immediately. sin37=0.6, cos37=0.8 given when needed.

6. Units on every line, diagrams for FBD/incline/pulley (he draws them in his notes), reasoning words: "because", "therefore", "since net force..." — his papers reward written justification over bare answers.

BLOCK 5/7 FULL — CIRCULAR MOTION WORKSHEET Q1–Q30 (ALL SOLVED — HIS BANK)

Q1. r=2 m, arc 6 m. θ=s/r=**3 rad**.

Q2. 120° → 120π/180=**2π/3 ≈ 2.09 rad**.

Q3. θ=3π/4, r=4 m. s=rθ=**3π ≈ 9.42 m**.

Q4. θ=60°, r=5 m. (x,y)=(r cos60, r sin60)=**(2.5, 4.33) m**.

Q5. r=3 m, v=6 m/s. $a_c=v^2/r=$**12 m/s²**.

Q6. r=2 m, ω=5 rad/s. (a) $v=r\omega=$**10 m/s** (b) $a_c=r\omega^2=$**50 m/s²**.

Q7. a_t=4 m/s², r=2 m. $\alpha=a_t/r=$**2 rad/s²**.

Q8. r=4, ω=3, α=2. (a) $a_t=r\alpha=$**8** (b) $a_r=r\omega^2=$**36** (c) $a=\sqrt{(64+1296)}=$**36.9 m/s²**.

Q9. 20 rev in 10 s. (a) $\theta=40\pi\approx$ **125.7 rad** (b) $\omega=$**4π ≈ 12.6 rad/s**.

Q10. T=0.5 s. $\omega=2\pi/T=$**4π ≈ 12.6 rad/s**.

Q11. f=5 Hz. (a) $\omega=2\pi f=$**10π ≈ 31.4 rad/s** (b) $T=1/f=$**0.2 s**.

Q12. 8π rad in 4 s. avg $\omega=$**2π ≈ 6.3 rad/s**; revs=8π/2π=**4**.

Q13. rest, α=3, t=5. (a) $\omega=\alpha t=$**15 rad/s** (b) $\theta=\frac{1}{2}\alpha t^2=$**37.5 rad**.

Q14. ω_0=4, α=3, t=6. ω=4+18=**22 rad/s**; θ=24+54=**78 rad**.

Q15. ω_0=20, α=−4→rest. (a) t=20/4=**5 s** (b) θ=(0−400)/(−8)=**50 rad**.

Q16. θ=20, 4→12 rad/s. α=(144−16)/40=**3.2 rad/s²**.

Q17. m=2, r=4, v=8. (a) $a_c=$**16 m/s²** (b) F=**32 N**.

Q18. 1000 kg, r=50, v=10. $F_c=1000\times100/50=$**2000 N**.

Q19. 5 kg, ω=4, r=2. (a) v=**8 m/s** (b) $a_c=$**32 m/s²** (c) $F_c=$**160 N**.

Q20 (explain — his style). Centripetal vs centrifugal. SCRIPT: Centripetal is the REAL inward net force (tension/friction/gravity) that continuously redirects the velocity of a body toward the centre so the circular path continues; it acts ON the moving body, in an inertial frame. Centrifugal is a PSEUDO (fictitious) force that appears only in the rotating (non-inertial) reference frame, directed radially outward; it has no physical agent/interaction — it accounts for the inertia of the body resisting the inward turn. Examples: centripetal — tension in the string of a whirled stone, friction force banking a car, gravity on a satellite; centrifugal — the outward push felt by a passenger on a merry-go-round, clothes in a spinning dryer flinging water outward.

Q21. 4 kg, L=2 m, v=6 m/s horizontal. $T=mv^2/r=$**72 N**.

Q22. 2 kg, 1.5 m, 4 m/s. (a) $a_c=16/1.5=$**10.7 m/s²** (b) T=**21.3 N**.

Q23. 1200 kg, r=40, v=12. $F_c=1200\times144/40=$**4320 N**.

Q24 (friction-circular combo!). 5 kg, r=2 m, μ=0.4, max v no-slip. $\mu mg=mv^2/r\rightarrow v=\sqrt{(\mu gr)}=\sqrt{7.84}=$**2.8 m/s**.

Q25. r=5, ω=2, α=3. (a) $a_t=$**15** (b) $a_r=$**20** (c) $a=\sqrt{(225+400)}=$**25 m/s²**.

Q26. 3 kg, r=2, 4→8 m/s in 2 s. (a) a_t avg=**2 m/s²** (b) a_c at 8: **32 m/s²** (c) $a=\sqrt{(4+1024)}=$**32.1 m/s²**.

Q27 (comment question!). r=100, v=20: $a_c=$**4 m/s²**. Speed doubled: $a_c'=v'^2/r=4v^2/r=$**16 m/s²**. COMMENT: centripetal acceleration depends on the SQUARE of speed — doubling v multiplies a_c by 4 (r fixed). This is why highway curves require much stricter speed limits for heavy/fast vehicles.

Q28. r=0.5, 120 rpm. (a) $\omega=$**4π ≈ 12.6 rad/s** (b) $v=$**2π ≈ 6.3 m/s** (c) $a_c=0.5\times16\pi^2=$**8π² ≈ 79 m/s²**.

Q29 (function type!). r=3, $\theta(t)=2t^2+3t$. (a) $\omega=d\theta/dt=4t+3\rightarrow t=2:$ **11 rad/s** (b) $\alpha=$**4 rad/s²** (c) $v=r\omega=$**33 m/s** (d) $a_t=r\alpha=$**12 m/s²** (e) $a_c=r\omega^2=$**363 m/s²**.

Q30. r=2, $\omega(t)=3t^2+2$. (a) $\alpha=6t\rightarrow t=2:$ **12 rad/s²** (b) $v=r\omega=2(14)=$**28 m/s** (c) $a_t=r\alpha=$**24 m/s²** (d) $a_c=r\omega^2=2\times196=$**392 m/s²**.

BLOCK 6 FULL — WORK & ENERGY WORKSHEET Q1–Q12 (SOLVED)

Q1. 5 kg, 4 m/s. $KE=\frac{1}{2}\times5\times16=$**40 J**.

Q2. p=10 kg·m/s, m=2. $K=p^2/2m=100/4=$**25 J**.

Q3. 5 kg, h=10 m. $PE=5\times9.8\times10=$**490 J**.

Q4. F=80 N, d=8 m along force. W=**640 J**.

Q5. F=100 N, d=5 m at 60°. W=100×5×cos60=**250 J**.

Q6. Spring 0.1→0.3 m, k=200. Work by spring = $-\frac{1}{2}k(x_2^2-x_1^2)=-100\times(0.09-0.01)=$**−8 J** (spring does negative work when stretched further; work done by external agent = +8 J).

Q7 (explain!). Same work both paths → FORCE IS CONSERVATIVE. SCRIPT: Work done is path-independent (equal for the two paths), hence the work over any closed loop is zero; that is exactly the defining property of a conservative force. (Contrast: friction would give longer path → more negative work — path-dependent, non-conservative.)

Q8. 4 kg rest, W_net=200 J. $200=\frac{1}{2}\times4\times v^2$ → v=**10 m/s**.

Q9. 5 kg, 4 m/s, W=90 J. $90=2.5(v^2-16)$ → $v=\sqrt{52}$=**7.21 m/s**.

Q10. 2 kg from rest h=20. $v=\sqrt{(2gh)}=\sqrt{392}$=**19.8 m/s**.

Q11. 5 kg h=8, friction −50 J. $392-50=2.5v^2$ → $v=\sqrt{136.8}$=**11.7 m/s**.

Q12. 10 kg h=5, friction −100 J. $490-100=5v^2$ → $v=\sqrt{78}$=**8.83 m/s**.

Q13 (5-part!). 4 kg pushed 30 N through 10 m, friction 5 N opposing, from rest. (a) W_app=30×10=**300 J** (b) W_fric=−5×10=**−50 J** (c) W_net=**250 J** (d) KE_f=250 J (e) $v=\sqrt{(2\times250/4)}=\sqrt{125}$=**11.2 m/s**.

Q14. $KE=\frac{1}{2}mv^2=K$ kept constant, m→4m. $\frac{1}{2}mv^2=\frac{1}{2}(4m)v'^2$ → **v'=v/2** (speed halves; momentum doubles... wait p=4m×v/2=2mv — increases).

Q15 (explain!). A→B with gravity+friction. SCRIPT: Gravity is conservative — its work (−mgh) depends only on the vertical height difference, not path. Friction is non-conservative — it always does negative work $-f_k\cdot d$ proportional to path length, converting mechanical energy into heat. Hence total mechanical energy (KE+PE) is NOT conserved between A and B: $\Delta E_{mech} = W_{friction} = -f_k\, d$. (Total energy including heat is conserved.)

BLOCK 8 FULL — COLLISIONS WORKSHEET Q1–Q12 (SOLVED)

Q1. 2 kg @6 m/s head-on, stationary 4 kg. Total momentum before = 2×6 = **12 kg·m/s**.

Q2. 5 kg @4 + stationary 3, stick. v=(5×4)/8=**2.5 m/s**.

Q3. 2 kg @8 + stationary 6, perfectly inelastic. (a) p_i=16 (b) v=16/8=**2 m/s** (c) $KE_i=\frac{1}{2}\times2\times64$=**64 J** (d) $KE_f=\frac{1}{2}\times8\times4$=**16 J** (e) lost=**48 J**. (Check: $\frac{1}{2}\mu u^2=\frac{1}{2}(12/8)\times64=48$ ✓.)

Q4. 2 kg @10 elastic, stationary 3. $v1=(m1-m2)u/(m1+m2)=-10/5=$**−2 m/s** (rebounds back); $v2=2m1u/(m1+m2)=40/5=$**8 m/s**.

Q5. 4 kg @6 vs 2 kg @3 opposite. u1=6, u2=−3. $v1=[(4-2)6+2\times2\times(-3)]/6=(12-12)/6=$**0**; $v2=[(2-4)(-3)+2\times4\times6]/6=(6+48)/6=$**9 m/s**.

Q6. 5 kg @8 + stationary 3, stick. (a) **5 m/s** (b) KE_i=160 J (c) KE_f=100 J (d) lost=**60 J**.

Q7. 2 kg @10, stationary 3, e=0.6. $v2-v1=6$; $2v1+3v2=20$ → v1=**0.4**, v2=**6.4 m/s**.

Q8. 4 kg @8, stationary 6, e=0.5. $v2-v1=4$; $4v1+6v2=32$ → v1=**0.8**, v2=**4.8 m/s**.

Q9. 3 kg @6, stationary 2, e=0.4. approach=6; separation=**2.4 m/s**; solve → v1=**2.64**, v2=**5.04 m/s**.

Q10. 2 kg @12, stationary 4, e=0.5. $v2-v1=6$; $2v1+4v2=24$ → v2=6, v1=**0**; v2=**6 m/s**. (Special: lighter object stops dead.)

Q11 (2D stick — HIS FAVOURITE). 4 kg east@5 + 6 kg north@3, stick. px=20, py=18. $v=\sqrt{(400+324)}/10=\sqrt{724}/10\approx$**2.69 m/s**; $\theta=\tan^{-1}(18/20)\approx$**42° north of east**.

Q12. 5 kg +x@8, 3 kg +y@6, stick. px=40, py=18. $v=\sqrt{(1600+324)}/8\approx$**5.48 m/s**; $\theta=\tan^{-1}(18/40)\approx$**24° from x-axis**.

Q13. Identical balls elastic, first@8, second at rest → **swap**: (0, 8 m/s).

Q9-rematch from photo Q15-type: 4 kg block at speed, e=0.5 — same two-equation machine.

KE-loss shortcut: $\Delta KE=\frac{1}{2}\, m1m2(u1-u2)^2(1-e^2)/(m1+m2)$. Perfectly inelastic: $\Delta KE=\frac{1}{2}\mu u^2$ (max possible loss).

HIS CONTEST-1 5M QUESTIONS (SOLVED — closest to Section B tomorrow)

Q21 error-propagation (memorize this exact template). F=80.0±0.8 N, a=25.0±0.5 m/s². (1) m=F/a=**3.20 kg** (2) Δm/m=0.8/80+0.5/25=0.03 → Δm=**0.096 kg** (3) relative=**0.03** (4) % = **3%**. RULE: multiply/divide → relative errors ADD; power n → multiply by n; add/subtract → absolute errors ADD.

a(t) integration. a=10t+6, v(0)=4, x(0)=3. v(t)=∫a=**5t²+6t+4**; x(t)=∫v=**(5/3)t³+3t²+4t+3**; x(1)=**14.67 m**. (His integration questions = this exact skeleton: integrate, fix constant from initial condition, evaluate.)

Car traffic-signal. 0→20 m/s in 5 s, then 20 for 5 s. Displacement=area under v-t=½×5×20+5×20=**100 m**; avg v=**10 m/s**; avg a first 5 s=**4 m/s²**. Graph: ramp then flat.

Q24 projectile (3-part). 25 m/s at 37°: ux=20, uy=15 (sin37=0.6, cos37=0.8 given); T=2uy/g=3.06 s (g=9.8) or 3.0 (g=10); H=uy²/2g=11.5 m (or 11.25).

Q28b airplane. 100 north + wind 60 east: |v|=√(10000+3600)=**116.6 km/h**; θ=tan⁻¹(60/100)=**31° east of north**. (Check if question asks "where must it point to go due north": then aim θ west of north with sinθ=60/100 → θ=37°; read the exact ask.)

HIS CONTEST-1 MCQs (1M each — definition precision)

Q3: unpredictable variations = **random error**. Q8: max range at **45°**. Q9: sig figs in 0.00456 = **3**. Q10: SI unit acceleration = **m/s²**. Q11: perpendicular vectors dot product = **5×4×cos90=0**. Q12: (30−10)/5=**4 m/s²**. Q13: negative scalar reverses **direction**.

MOMENTUM WORKSHEET Q5–Q9 (SOLVED)

Q5. Rod 6 m, 12 kg, from x=2 to x=8. CoM=midpoint=**x=5 m**.

Q6. 5 kg @4. p=**20 kg·m/s**.

Q7. 2 kg right@8 + 4 kg left@3. Total p=16−12=**4 kg·m/s to the right**.

Q8 (wall-rebound classic). 0.5 kg @10 hits wall, rebounds @6 opposite. (a) p_i=5 (b) p_f=−3 (c) Δp=**−8 kg·m/s** (d) impulse=**−8 N·s** (direction: away from wall... actually Δp is opposite to initial direction = outward).

Q9. 4 kg @6 + stationary 2, stick. v=24/6=**4 m/s**.

CoM WORKSHEET Q1–Q4 (SOLVED)

Q1. 2 kg at 0, 8 kg at 5. x_CM=(8×5)/10=**4 m**.

Q2. (2,0),(3,4),(5,10). x_CM=(0+12+50)/10=**6.2 m**.

Q3. 2 & 8 kg, 10 m apart. From 2 kg: x=(8×10)/10=**8 m** (near the heavy one — CoM always closer to larger mass).

Q4. 2@(0,0), 3@(6,0), 5@(0,4). x=18/10=**1.8**; y=20/10=**2.0** → **(1.8, 2.0)**. (His student note: solve x and y separately → (x,y).)

THE 30-SECOND TEMPLATES FOR HIS FAVOURITE 5M SHAPES

1. **Error propagation:** m=F/a style → value, Δ via relative add, relative, %.
2. **a(t)→v(t)→x(t):** integrate twice, pin constants, evaluate.
3. **v-t graph story:** draw ramp/flat, displacement=area, avg v=disp/time, avg a=Δv/Δt.
4. **Projectile (a)(b)(c):** components → T → H. (Add R if asked.)
5. **Collision with e:** momentum eq + restitution eq → solve system.
6. **2D stick:** conserve px and py separately → $v=\sqrt{(vx^2+vy^2)}$, $\theta=\tan^{-1}(py/px)$.
7. **Incline+pulley friction:** whole-system a=(driving−resisting)/total mass; T from one body.
8. **CoM 2D:** weighted means per axis.
9. **Ice-tray explain:** claim → redistribution → direction → mass-ratio controls magnitude.

CONTEST-1 Q18–Q20 + FULL Q28b (VERBATIM, SOLVED)

Q18 (2). dy/dx of $y=4x^2-3x+7$ → **$8x-3$**. $\int(3x^2+2x)dx$ → **x^3+x^2+C**.

Q19 (2). Graph 1 (straight line, rising): uniform motion — velocity constant; avg velocity = instantaneous velocity = slope (both constant). Graph 2 (curving upward, slope steepens): velocity increasing with time; instantaneous = slope grows; avg over [0,t] also grows.

Q20 (2). A=20 m/s, B=12 m/s same dir: $v_{A/B}$=**8 m/s**. Opposite: **32 m/s**.

Q28b (10) An Airplane Flying in Wind. Plane 100 km/h through air, points due north; wind 60 km/h east. East=+î, north=+ĵ.

(a) Pilot's claim WRONG: ground velocity = sum of air-relative + wind; pointing north still adds eastward wind → drifts northeast.

(b) $v_{air}=100\hat{j}$, $v_{wind}=60\hat{i}$ → v_{ground}=**$60\hat{i}+100\hat{j}$**, $|v|=\sqrt{(3600+10000)}$=**116.6 km/h**, θ=**31° east of north**. In 30 min: displacement = **$50\hat{i}+50\hat{j}$ km** (30E, 50N... actually (30,50)). Sketch vector triangle (north leg 100, east leg 60, hypotenuse).

(c) To go due north: east component of v_{air} must cancel wind: $v_{air,x}$ = **–60 km/h** (i.e., 60 west), north component = **100 km/h** (full? no: speed fixed at 100 → north comp = $\sqrt{(10000-3600)}$=**80 km/h**). Ground speed = 80 km/h due north. Point slightly west of north (sketch only, no angle).

(d) Wind 120 > plane speed 100: CANNOT — east component can cancel at most 100, wind is 120. Impossible.

MOMENTUM WORKSHEET Q10–Q17 (SOLVED)

Q10. 2 kg east@6, 3 kg north@4. (a) p_x=**12** (b) p_y=**12** (c) |p|=**$12\sqrt{2}\approx17.0$ kg·m/s** (d) $\theta=\tan^{-1}(12/12)$=**45° north of east**.

Q11. 2 kg east@5 + 1 kg north@4, stick. px=10, py=4 → vx=10/3, vy=4/3. (b) $v=\sqrt{(100+16)}/3$=**$\sqrt{116}/3\approx3.57$ m/s** (c) $\theta=\tan^{-1}(4/10)$=**21.8° north of east**.

Q12. 2 kg@4 + 3 kg@6 same dir. (a) p=8+18=**26** (b) v_{cm}=26/5=**5.2 m/s**.

Q13. Mass 20, F=60 N → a_{cm}=**3 m/s²**. (His note: F=Ma on the whole system.)

Q14. m=10, a_{cm}=2 → F_{net}=**20 N**.

Q15 (double collision). 2@10 → hit 3 stationary stick → v=4; then 5 kg total @4 hits 5 stationary stick → **v=2 m/s**.

Q16 (explosion). 10 kg at rest → 4 kg piece @12. 0=48+6v → **v=–8 m/s** (6 kg piece: 8 m/s opposite).

Q17. F_net=0: (a) total momentum **constant/conserved** (b) v_cm **constant** (c) a_cm=**0**.

CONTEST-1 Q28a (10M), Q24, Q25 (FULL, SOLVED)

Q28a Ball-in-accelerating-truck (g=10). Truck 10 m/s, a=2; ball thrown 5.0 m/s up relative to truck.

(a) Claim WRONG: ball keeps truck's release velocity, but truck accelerates → truck pulls ahead.

(b) v_ball ground=10î+5ĵ. T=2×5/10=**1.0 s**. Ball x=10 m; hand x=10+½·2·1=11 m → **1 m behind the hand**.

(c) Need ball x = hand x over T=1 s: vx·1=11 → vx=11 ground → **+1 m/s relative to truck (forward)**. Doubling truck's initial speed: truck disp=20·1+1=21, ball needs vx=21 → rel=21–20=**still 1 m/s** — design UNCHANGED (relative toss depends only on ½aT, not u).

(d) Time 1.0 s from 5.0 and 10.0 (2 sig figs) → **2 significant digits**.

Q24 Projectile (g=10 GIVEN, sin37=0.6, cos37=0.8). 25 m/s @37°: ux=**20**, uy=**15**; T=2uy/g=**3.0 s**; H=$uy^2/2g$=**11.25 m**. (Check g per question: Q24 says g=10; worksheets say g=9.8 — READ IT.)

Q25 Drone s(t)/y(t). x=$3t^2+4t+2$, y=$2t^3-3t^2+5t$. vx=6t+4, vy=$6t^2-6t+5$. t=2: vx=**16**, vy=**17**, speed=**√545≈23.3 m/s**. ax=6, ay=12t–6 → t=2: ay=**18**, |a|=**√360≈19.0 m/s²**. θ=$\tan^{-1}(17/16)$=**46.7°** above +x.

FINAL EXAM-DAY REMINDER (physics-setter's DNA)

His Part-D 10M: "Evaluate a claim → Support your judgment (compute) → Design → Check/Assess a limitation" — ALWAYS answer all four verb-led parts in order, even if qualitative. His Part-C 5M: clean multi-part calc with given g and trig values. His Part-B 2M: one derivative/integral/graph statement. His Part-A 1M: exact-definition MCQ. If a mock appears tomorrow, expect sections in that order. Formula card: p=mv; $p_i=p_f$; e=$-v_{rel(f)}/v_{rel(i)}$; $v_{cm}=\Sigma m_i v_i/\Sigma m_i$; $a_{cm}=F_{ext}/M$; $x_{cm}=\Sigma m_i x_i/M$; KE=½mv^2; W=∫F·dr; work-energy ΔKE=W; friction f=μN (kinetic), f≤μsN; $v^2=u^2+2as$; T=2π√(L/g), $\omega^2=r$, $a_c=v^2/r=\omega^2 r$; relative v A/B=vA–vB; error: mult→relative add, power→×n.

CONTEST-1 PART-A/B MCQs + SHORTS (FULL, SOLVED)

Q3. Unpredictable variations → **random error**. Q4. v=u+at=0+2(5)=**10 m/s**. Q5. Vector → **velocity**. Q6. Cross-product direction → **right-hand rule** (NOT parallelogram). Q7. Rate of change of velocity → **acceleration** (jerk = rate of change of acceleration — trap).

Q13. Vector × negative scalar → **direction reversed**. Q14. Triangle law → vectors arranged **head-to-tail**. Q15. Projectile vertical acceleration → **–g** (constant, downward; NOT zero, NOT +g).

Q16 (2). Order of magnitude. Write x=a×10^n with 1≤a<10; if a≥√10≈3.16 round n up. 0.00000487=4.87×10^{-6}, a=4.87≥3.16 → **10^{-5}**. 564000000=5.64×10^8, a≥3.16 → **10^9**.

Q17 (2). Dot vs cross. Dot: scalar result, A·B=ABcosθ, commutative, defined via angle only. Cross: vector result (direction by right-hand rule, ⊥ to both), A×B=ABsinθ n̂, anti-commutative (A×B=−B×A). At least 2 differences each.